SPICA TIDE

DECK CARGO PLANNER

PSV FAR SPICA · NEO Energy Resources UK · North Sea Operations

12 Bays · 53.7m Deck Length · 2,500T Max Capacity · 7 Platforms

Version 1.0.0 · Build 38.20 · Beta

⚓ OVERVIEW

SPICA TIDE is a professional deck cargo planning application built for Platform Supply Vessel **FAR SPICA** operating in the North Sea under NEO Energy. It provides an interactive drag-and-drop deck plan with real-time cargo visualisation, platform colour coding, DG safety checks, and full PDF/Excel manifest export.

The application replaces paper-based and spreadsheet-based cargo plans with a precise, scalable digital tool that officers can use at the planning desk or on a bridge tablet.

🚢 12 Cargo Bays

Full deck from Bay 12 (aft) to Bay 1 (bow), each individually labelled and sized to real vessel dimensions.

⚖ 7 Platforms

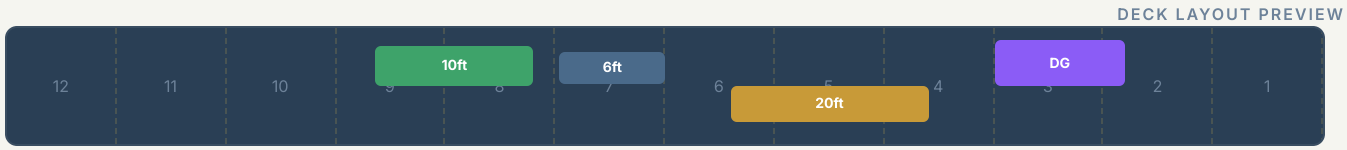
Bleo Holm, Claymore CAP/CPP/WOPS/Drill, Piper, Saltire — each with unique colour coding and cargo tracking.

⚠ DG Safety

Full IMDG segregation matrix with 20 DG classes. Automatic pairwise conflict detection with visual warnings.

📄 Export Ready

Professional PDF deck plans and Excel manifests exported directly to disk via native save dialogs.



⚙ CORE FEATURES

👁 Drag & Drop Placement

Select cargo from the Library panel and drag it directly onto the deck canvas. Drop to place at the exact position. Cargo blocks can be repositioned by dragging, rotated 90° with the R key, duplicated with D, or deleted with Del.

📚 Cargo Library Panel

The bottom panel contains 60+ preset cargo types across 5 categories: Container, Module, Tank, Basket, Skip. Each preset includes real-world dimensions in metres and default weights. Custom cargo types can be added and saved.

🎨 Platform Colour Coding

Each platform destination is assigned a unique colour. Cargo blocks are tinted by their assigned platform, making it easy to visually identify which cargo goes where. Colours can be manually changed by clicking the colour dot on any location card.

📌 Cargo Statuses

Four cargo statuses drive colour tinting and manifest grouping:

■ Load

Cargo being loaded onto vessel for delivery

■ Backload

Cargo being returned from platform to shore

■ ROB

Remaining On Board from previous voyage

■ Transfer

Cargo transferring between platforms (dashed border)

📄 ASCO / iLMS Excel Import

Upload an ASCO loadout Excel file (.xlsx) and the app parses cargo descriptions, CCU IDs, weights, DG classes, dimensions, and platform assignments automatically. Imported items appear in a queue for selective placement.

⚠ DG Dangerous Goods

20 IMDG DG classes with a full segregation matrix. The system automatically checks every cargo pair for DG violations. Conflicts show a pulsing red outline and a modal warning. Each conflict can be individually acknowledged.

📄 PDF Export

Generates a professional A4 landscape PDF with header, voyage info, stats, location cards, DG summary, full deck image, bay labels, voyage notes, and footer. Opens native Save As dialog — user chooses destination.

📊 Excel Export

Two-sheet Excel workbook: Cargo Manifest (all items with CCU, dimensions, weight, status, platform, DG) and Summary (totals, per-location breakdown, voyage notes). Native Save As dialog.

↰ Undo / Redo

50-step undo history. Every cargo mutation (move, add, delete, edit, rotate, duplicate) is tracked. Ctrl+Z to undo, Ctrl+Y or Ctrl+Shift+Z to redo. UI buttons also available in the header.

💾 Save / Open Project

Save the complete deck state as a .json project file. Open previously saved projects. Ctrl+S saves to current file (or opens Save As if new). Autosave to localStorage every 15 seconds as crash protection.

🔍 Zoom Controls

Zoom 30%–200% with smooth transitions. Ctrl+Scroll for wheel zoom. Number keys 1–5 for preset levels. Fit-to-screen button. Pinch-to-zoom on touch devices.

🌙 Dark / Light Theme

Full dark mode with warm graphite surfaces. Toggle via the theme pill in the header. All UI elements, cargo blocks, modals, and export dialogs are dark-mode aware.

⚙ SMART TOOLS PANEL

Access via the gear icon in the header or View menu. All settings persist between sessions.

🚀 **Smart Bounce**
Resolves overlapping cargo after drop by pushing to nearest free space
ON

📌 **Smart Grid Snap**
Snaps dropped cargo to nearest bay line, neighbour edge, or centre line
ON

⚠ **DG Auto-Segregation**
Checks every DG pair against IMDG segregation matrix. Safety critical.
ON

🔢 **Weight Gauge**
Animated fill bar in header showing total weight vs 2500T capacity
OFF

⚠ **Show DG Only**
Hides all non-DG cargo for quick compliance review
OFF

🔎 **Highlight by Platform**
Click a platform card to dim all other platforms' cargo to 25% opacity
OFF

📄 **Empty Deck Hint**
Shows placement hint icon when deck has zero cargo
ON

🎨 VISUAL EFFECTS

All visual effects are individually toggleable in the Visual Effects section of Smart Tools. Each persists between sessions.

🔷 **Corner Badges**
Small triangular badge on cargo block corners showing DG class or BL/ROB status
OFF

📈 **Animated Counter**
Airport departure-board style number animation when importing ASCO files
OFF

➖ **Bay Dashed Lines**
Yellow dashed lines replacing grey bay dividers — styled like real deck markings
OFF

✨ **Cargo Hover Glow**
Soft glow in platform colour around cargo blocks on mouse hover
OFF

⚓ **PORT / STBD Labels**
Vertical nautical side labels on both edges of the deck canvas
ON

🔢 **Section Watermarks**
Large semi-transparent bay numbers as background inside each section
ON

👻 **Drag Ghost Trail**
Semi-transparent ghost remains at original position while dragging cargo
OFF

🌈 **Smooth Colours**
Animated CSS transitions when cargo block platform/colour changes
ON

⚙ **Status Bar Icons**
Small SVG icons next to values in the bottom status bar
OFF

✨ **Name Shimmer**
Subtle shine animation on the SPICA TIDE vessel name in the header
OFF

👉 **Button Effects**
Micro press-down interaction on all buttons (1px translate + shadow)
ON

👁️ **Deck Shadow**

Drop shadow beneath deck plan creating physical depth illusion

ON

📜 **Custom Scrollbars**

Theme-coloured scrollbars in the library panel matching app accent

ON

KEYBOARD SHORTCUTS

KEY	ACTION
E	Edit selected cargo block
R	Rotate selected block 90°
D	Duplicate selected block
Del	Delete selected block
1 – 5	Zoom to 50% / 75% / 100% / 125% / 150%
L	Toggle library panel collapse
?	Show keyboard shortcuts cheatsheet
Esc	Deselect / cancel / clear filter
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + S	Save project
Ctrl + Shift + S	Save As (new file)
Ctrl + O	Open project
↑↓←→	Move selected block 1px
Shift + Arrow	Move selected block 1 metre
Ctrl + Scroll	Zoom in/out

📋 WORKFLOW GUIDE

A typical voyage planning workflow from start to PDF export:

1. **Set Voyage Details**
   Enter the voyage number and select the date in the header. Add voyage notes if needed.

2. **Activate Platforms**
   Click the ⊕ button in the location strip to add the platforms this voyage will visit. Each gets a unique colour.

3. **Import ASCO File (Optional)**
   Upload the ASCO/iLMS Excel loadout file. Review imported items in the queue. Select items to place on deck.

4. **Place Cargo**
   Drag cargo from the Library panel onto the deck, or click a library card then click the deck to place. Position, rotate (R), and resize as needed.

5. **Assign Status & Platform**
   Click each cargo block to open the edit modal. Set the correct status (Load/BL/ROB/TR), platform, weight, and DG class.

6. **Review DG Compliance**
   Check for DG segregation warnings (red pulsing outlines). Acknowledge or resolve each conflict.

7. **Export PDF**
   File → Export PDF. Choose destination folder and filename in the Save As dialog. The PDF includes full deck image, stats, and manifest data.

8. **Export Excel**
   File → Export Excel for the formal cargo manifest spreadsheet with per-location breakdown.

9. **Save Project**
   Ctrl+S to save the project file for future reference or continuation.

💡 TIPS & BEST PRACTICES

★ Use Smart Grid Snap

Keep Smart Grid Snap enabled (default ON). It aligns cargo to bay lines and neighbours automatically, keeping the deck plan clean and realistic.

★ Zoom to 100% Before Export

For the best PDF quality, set zoom to 100% before exporting. The deck image is captured at the current view.

★ Use Platform Highlight

Enable "Highlight by Platform" in Smart Tools, then click a platform card to isolate its cargo visually. Great for verifying platform-specific loads.

★ Save Early, Save Often

Autosave protects against crashes (every 15s to localStorage), but Ctrl+S creates a proper named project file you can share or archive.

★ Right-Click Context Menu

Right-click any cargo block for quick access to Edit, Rotate, Duplicate, and Delete actions.

★ Keyboard Shortcuts Speed

Learn E/R/D/Del for cargo editing — much faster than using the mouse buttons. Arrow keys for precision positioning (Shift+Arrow = 1 metre).

★ DG Acknowledgement

DG warnings are per-cargo-pair, not per-class. Acknowledging a warning for two specific containers won't suppress future warnings for new pairs of the same DG class.

© 2026 SPICA TIDE Marine Operations

Version 1.0.0 · Build 38.20